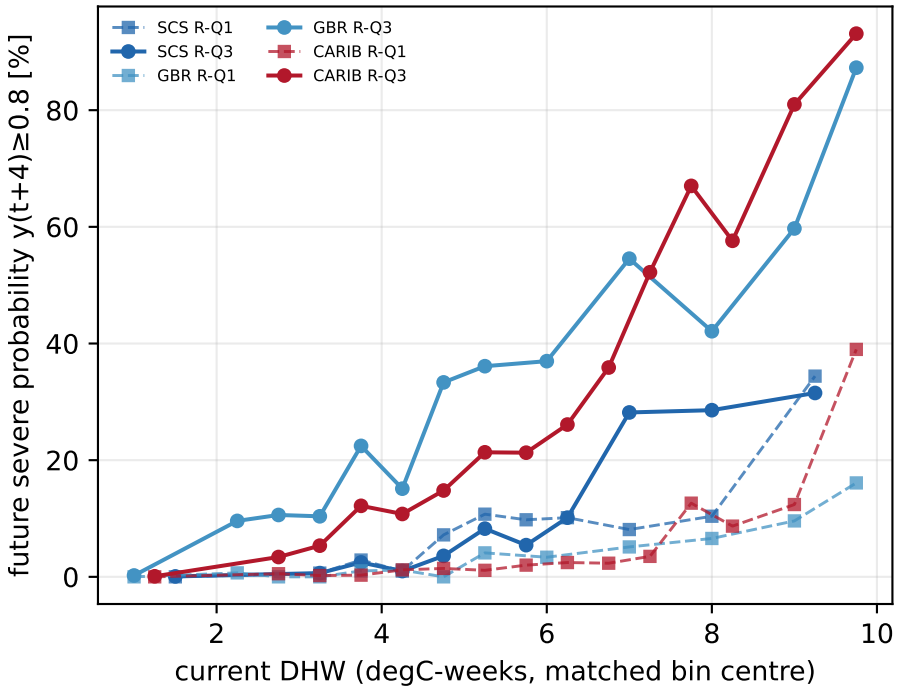
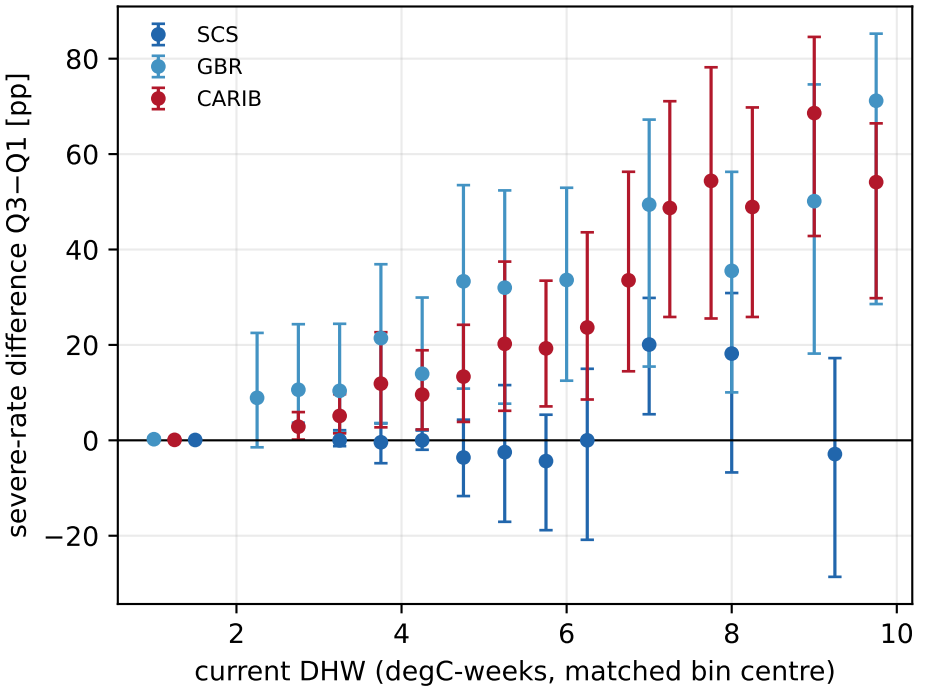
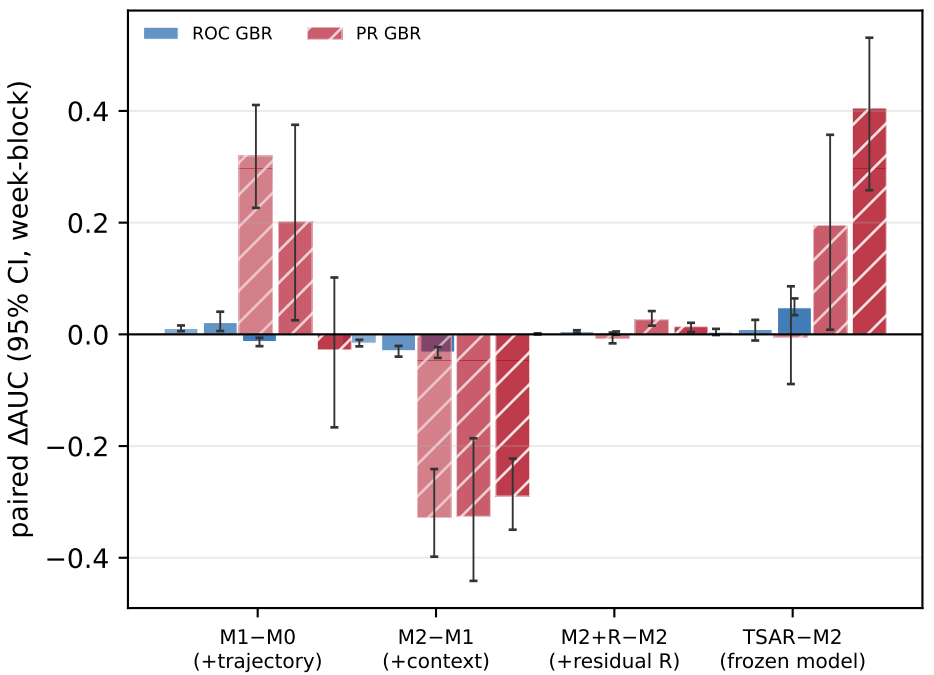
a DHW-matched response: R-Q1 vs R-Q3



b Within-DHW residual risk gradient (95% CI)



c Incremental ladder M0→M1→M2, +R, →TSAR



d Evidence matrix (target basins GBR / CARIB; E3 gradient-diff CI [-0.022, +0.431] incl 0)

Basin	E1 partial ρ CI excl 0	TSAR−M2 ΔROC	E5 M2+R gain
SCS	no	+nan	no
GBR	YES	+0.0089	YES
CARIB	YES	+0.0477	YES

Frozen records; R = TSAR(seed-42) − PersA; severe y(t+4)≥0.8; week-block bootstrap 1000 seed 42; associative / representation-level only — no causal claim.